

TOP-K IS NOT A BUDGET FOR HYBRID RETRIEVAL

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ABSTRACT

Hybrid retrieval for RAG often fuses fixed-depth dense and sparse rankings. Exact fusion avoids this truncation, but completing a fixed Top- K can require sharply different numbers of accesses across queries. DiBud (Direct Budgeting) takes an access budget directly and returns the certified prefix available within it. Incremental certification determines the output size, while selective access resolves remaining ranking uncertainty. Replays on five query sets show that, at 2048 accesses, DiBud certifies 7.86% more results within the first 100 positions than balanced access. After budget calibration for 95% of exact Top-20 quality, held-out NFCorpus and SciFact queries retain 95.81% and 97.68% of nDCG@20 while reducing accesses by 65.92% and 83.41%. Complete Top-20 measurements reveal long-tailed access costs, while five corpus snapshots show how output size varies under a fixed budget.

Index Terms— Hybrid retrieval, reciprocal rank fusion, budgeted retrieval, exact prefix, RAG

1. INTRODUCTION

Hybrid retrieval for retrieval-augmented generation (RAG) commonly combines the top L results from dense and sparse retrievers using reciprocal rank fusion (RRF) [1, 2]. A fixed window limits the number of entries accessed but also excludes contributions beyond that window. When a document appears in one ranked list but not in the other list’s window, fusion assigns zero to the latter contribution. Yet absence from the window only means that the rank has not been observed; its contribution remains unknown. Replacing an unknown contribution with zero can change the output order even when the candidate set already contains every document in the top K under RRF over the full ranked lists. Thus, L is a retrieval-depth parameter that affects both retrieval quality and access cost.

In practice, L can be selected using historical queries to balance quality and cost. However, RAG systems serving open-ended requests over evolving knowledge bases continually encounter new queries and updated corpora. Both

changes alter the ranked lists and the positions of cross-channel contributions, making it difficult for a previously selected L to preserve the original quality–cost tradeoff. Exact rank aggregation maintains bounds on unobserved contributions and accesses the ranked lists until the required results are determined [3]. EAHR applies this approach to hybrid retrieval to obtain the exact top K in order without a predefined truncation depth [4]. Nevertheless, the number of accesses required can vary sharply across queries: resolving competition at just a few positions can require accessing entries far down the ranked lists.

Because the same K can require substantially different numbers of accesses, resource-constrained requests need to bound accesses directly rather than control cost indirectly through K . We take an access budget as input, defined as the maximum total number of sorted accesses to the two ranked lists, without specifying the output size in advance. Each sorted access retrieves the next entry from one list. The system incrementally certifies results and returns a certified prefix when the budget is exhausted: an initial segment guaranteed to match the RRF ranking over the full lists in both membership and order. Differences in query difficulty therefore translate into different output sizes, rather than additional accesses required to complete a fixed top- K result.

To support retrieval under an access budget, we introduce DiBud (Direct Budgeting). DiBud incrementally computes the fused ranking, appending results whose ranks have been certified [5] and directing subsequent accesses toward unresolved competition at later positions [6]. When the budget is exhausted, it returns the current certified prefix. No K is specified in advance; the output size is determined by the ranking information obtained within the budget.

We evaluate this approach through replays on five query sets and five corpus snapshots. We first examine fixed-depth transfer, then measure the access cost and incremental certification of exact top- K results. Under equal budgets, we compare DiBud with balanced sorted access, including output distributions and changes across snapshots. Finally, calibration on disjoint queries quantifies the relevance retained and accesses saved relative to completing exact Top-20.

2. DIBUD: BUDGETED EXACT-PREFIX FUSION

Given a query, a corpus snapshot, and a read budget B , DiBud reads the dense and sparse rankings on demand and returns a certified prefix of their complete RRF ranking. The budget determines how much can be read; certification determines how much can be returned.

2.1. Budget and output

Let $r_i(x)$ denote the one-based rank of document x in channel $i \in \{D, S\}$, and let $g(r) = 1/(c+r)$ for rank constant $c \geq 0$. Set $r_i(x) = \infty$ and $g(\infty) = 0$ if x is outside that channel's support. The complete fusion score is

$$F(x) = \sum_{i \in \{D, S\}} g(r_i(x)). \quad (1)$$

Sorting the union of the channel supports by decreasing $F(x)$, with a fixed document-identifier order to break ties, defines the complete sequence T^* .

Both channels supply resumable exact ranking prefixes from the same snapshot. If d_D and d_S are their read depths and A is the output sequence, the required contract is

$$d_D + d_S \leq B, \quad A = T_{1:K}^*, \quad K = |A|. \quad (2)$$

K is not specified in advance. The output is empty if no position can be certified. The budget counts channel entries read, including separate reads of the same document in different channels.

2.2. Prefix certification

Let S_i be the documents already read from channel i . An unread contribution in that channel is bounded by

$$u_i = \begin{cases} 0, & \text{if confirmed exhausted,} \\ g(d_i + 1), & \text{otherwise.} \end{cases} \quad (3)$$

For an observed document x , the score bounds are

$$\ell(x) = \sum_{i: x \in S_i} g(r_i(x)), \quad (4)$$

$$h(x) = \ell(x) + \sum_{i: x \notin S_i} u_i. \quad (5)$$

A document unseen in both channels has score at most $h_\emptyset = u_D + u_S$.

Among observed documents not yet emitted, select x with the highest lower bound, breaking ties by the fixed identifier order. Certify x as the next output when its lower bound exceeds every remaining competitor's upper bound and $h_\emptyset$. Equality with an observed competitor is allowed only when x wins the identifier tie; comparison with the unseen bound

remains strict. Append x and repeat until the next position cannot be certified.

The bounds ensure $\ell(x) \leq F(x) \leq h(x)$. Each emitted document therefore precedes every remaining document, including any not yet observed. Applying this argument at each output position establishes the prefix equality in (2).

2.3. Reading and stopping

When the next position cannot be certified, let y be the observed competitor with the highest upper bound, excluding x and the emitted prefix and breaking ties by identifier. If $h(y) \geq h_\emptyset$ and y is missing a contribution from exactly one channel, advance that channel. This either reveals the contribution or tightens its upper bound.

Otherwise, advance the channel whose next batch offers the larger reduction in the unread bound. For a batch of b entries in a channel that remains open after the read, this reduction is

$$\Delta_i = g(d_i + 1) - g(d_i + b + 1). \quad (6)$$

If exhaustion is established, the new bound is zero. Exhausted channels are skipped, and equal reductions are resolved by a fixed channel order (dense first). This rule also initializes reading when no document has been observed.

After each batch, update the bounds and extend the certified prefix. The batch size is capped by the remaining budget. When the budget is consumed, or both channels are exhausted, return the accumulated output. Certification depends on the observed bounds, so changing the read schedule affects the number of certified results without changing the meaning of an emitted position.

3. EXPERIMENTS

3.1. Setup

We evaluate 770 queries from five query sets—TREC-DL 2019/2020, NFCorpus, SciFact, and TREC-COVID—and the same 30 queries across five TREC-COVID snapshots [7, 8, 9, 10, 11]. We replay exact rankings from bge-small-en-v1.5 [12] and BM25 [13] ($k_1 = 0.9$, $b = 0.4$), using equal-weight RRF with contributions $1/(59 + r)$ and deterministic tie breaking.

First, we choose L on calibration queries and test it on held-out queries and later corpus snapshots. Next, we measure how many accesses are needed to complete exact Top- k and when each result becomes certain. We then compare DiBud with Balanced under the same budgets and certification rules. Balanced alternates between the two lists. Budgets range from 128 to 10000, and we count certified results up to 20 and 100. Both methods access one entry at a time; we also test batches of 64.

Table 1. Accesses to complete exact Top-20 under unit-access DiBud. All 770 queries complete.

Query set	n	Median	P95	Maximum
TREC-DL 2019	43	311	196410	3533518
TREC-DL 2020	54	444	35562	211169
NFCorpus	323	308	2685	5035
SciFact	300	325	3362	5011
TREC-COVID	50	2544	56363	91290

Finally, we compare budgeted stopping with completing exact Top-20. Budgets are selected on calibration queries to retain 90%, 95%, or 99% of the reference mean nDCG@20. We report the quality retained and accesses saved on held-out queries.

Results are averaged within each query set, then equally across sets. Where reported, 95% confidence intervals use 2000 query bootstrap samples. The static evaluation uses reconstructed full channel rankings: all 770 queries complete exact Top-20, and all tested budget endpoints are observed. The temporal evaluation retains the frozen snapshot rankings; saved-list boundaries are not treated as exhaustion.

3.2. Transfer of fixed depths

We evaluate $L \in \{10, 20, 50, 100, 200, 500, 1000, 2000, 5000\}$. On the same MS MARCO corpus, increasing L from 20 to 5000 raises nDCG@10 from 0.6538 to 0.6812 for TREC-DL 2019, but lowers it from 0.6424 to 0.6243 for TREC-DL 2020 (Fig. 1a). Changing queries reverses the benefit of a deeper window.

For the same 30 queries across five TREC-COVID snapshots, the best tested depths are 50, 100, 200, 200, and 5000 (Fig. 1b). We then test whether an earlier selection transfers: five-fold calibration selects L using Round-1 queries and freezes it for held-out queries across all rounds. The nDCG@10 difference from full RRF changes from -0.0085 in Round 1 to -0.0272 in Round 5 (95% CI: $[-0.0699, -0.0058]$; Fig. 1c). A depth selected on earlier data does not necessarily maintain its relative effectiveness as the corpus changes.

3.3. Fixed-K access cost

A fixed result count does not stabilize access cost (Table 1). On TREC-DL 2019, completing exact Top-20 takes a median of 311 accesses, but the P95 is 196410 and the maximum is 3533518: approximately 632 and 11362 times the median. The P95 is also 80 times the median on TREC-DL 2020 and 22 times on TREC-COVID. These differences motivate controlling accesses directly through a budget rather than indirectly through K .

Table 2. Mean certified output at $B = 2048$. Both methods use unit accesses and the same certifier. All endpoints in this table are observed.

Query set	Cap 100		Cap 20	
	Balanced	DiBud	Balanced	DiBud
TREC-DL 2019	41.86	46.19	17.77	17.91
TREC-DL 2020	34.91	38.31	18.33	18.35
NFCorpus	62.06	63.71	19.24	19.28
SciFact	43.55	47.26	19.44	19.57
TREC-COVID	26.48	29.80	15.58	15.78

Table 3. Held-out results after 95% budget calibration. Retention is the ratio of mean nDCG@20; savings compare total accesses with completed exact Top-20.

Query set	Retained (%)	Fewer accesses (%)
TREC-DL 2019	95.05	99.53
TREC-DL 2020	95.89	93.50
NFCorpus	95.81	65.92
SciFact	97.68	83.41
TREC-COVID	95.85	69.71

3.4. Certified output under equal budgets

At $B = 2048$, DiBud increases mean certified output within the first 100 positions from 41.77 to 45.05, a 7.86% gain over Balanced (Table 2). Within the first 20 positions, the mean changes from 18.07 to 18.18. All five sets gain output, with larger improvements for longer prefixes. Over the full log-budget range, DiBud has a larger normalized certified-prefix area than Balanced on all five static sets. The paired 95% confidence intervals for these differences remain above zero with both unit accesses and batches of 64.

Across the five snapshots at $B = 2048$, DiBud returns means of 18.00, 17.57, 16.40, 15.77, and 15.10 results within the first 20 positions, versus 17.70, 17.20, 16.03, 15.30, and 14.83 for Balanced. A fixed budget yields different output sizes as the corpus changes. DiBud improves the mean in every round. Full budget curves, output quantiles, empty-output rates, and delivery thresholds are retained with the experiment artifacts.

3.5. Quality and access cost relative to exact Top-20

We compare two stopping conditions on the same unit-SNRA trajectory: complete exact Top-20, or stop at the budget or at 20 certified results, whichever comes first. Evaluation uses linear-gain nDCG@20, with absent positions contributing zero. For each query set, five-fold calibration chooses the smallest budget retaining 90%, 95%, or 99% of the reference mean nDCG@20; the frozen choice is evaluated on held-out queries. Retention is a ratio of means, not a mean of per-query ratios with an arbitrary value for zero-reference queries. At

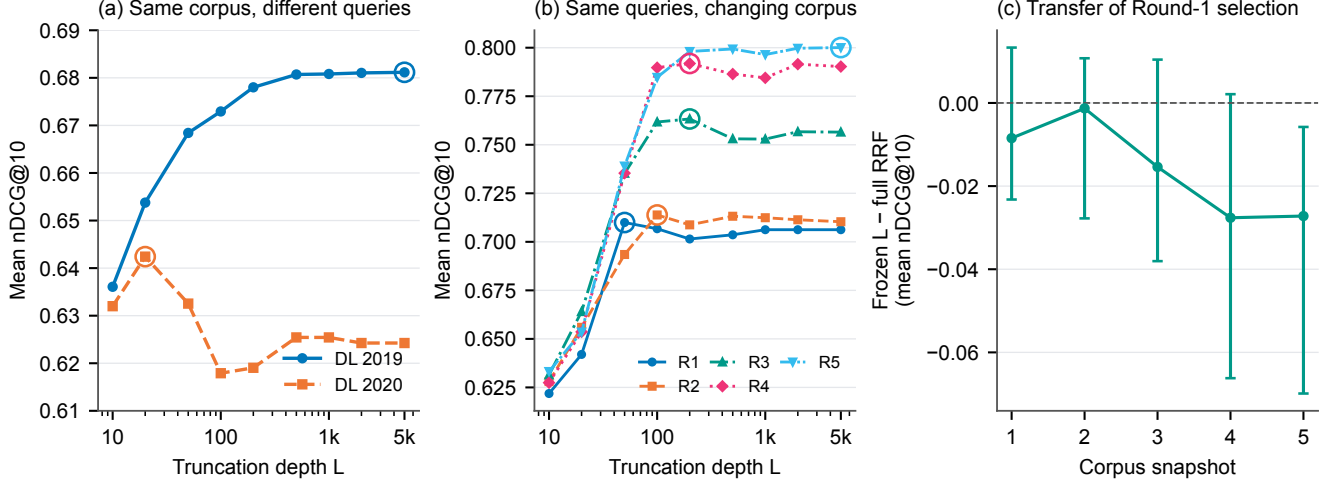


Fig. 1. Depth sensitivity and transfer. (a) Different queries on the same corpus. (b) The same queries across corpus snapshots, evaluated with each round’s judgments. Circled points mark retrospectively selected best depths. (c) Round-1 depths applied to held-out queries across rounds. Error bars show nested bootstrap 95% confidence intervals, including depth reselection and preserving query identities across rounds.

the 95% calibration target, held-out quality retention ranges from 95.05% to 97.68%, with 65.92%–99.53% fewer accesses across the five sets (Table 3; Fig. 3). TREC-DL 2019 retains 95.05% of mean nDCG@20 while saving 99.53% of accesses. Its large reduction reflects the cost of completing the long-tail queries in Section 3.3. Budgeted stopping preserves most of the measured relevance gain while avoiding much of this completion cost. At the 99% target, five, four, and five calibration folds on TREC-DL 2019, TREC-DL 2020, and TREC-COVID do not reach the target within the tested budgets. Higher quality targets can require budgets beyond the evaluated range.

4. CONCLUSION

Fixed-depth fusion can change its relative effectiveness across queries and corpus snapshots, while exact Top- K completion retains a variable access cost. DiBud directly bounds sorted accesses and lets incremental certification determine the returned prefix. Matched-budget replays show increased output over balanced access, and held-out budget calibration demonstrates substantial access savings at high mean quality retention on several query sets. The tradeoff varies across collections: fewer accesses can mean fewer returned results, and a quality target need not transfer to every query. The evidence concerns logical accesses to exact ranked lists; online retrieval latency remains outside this evaluation.

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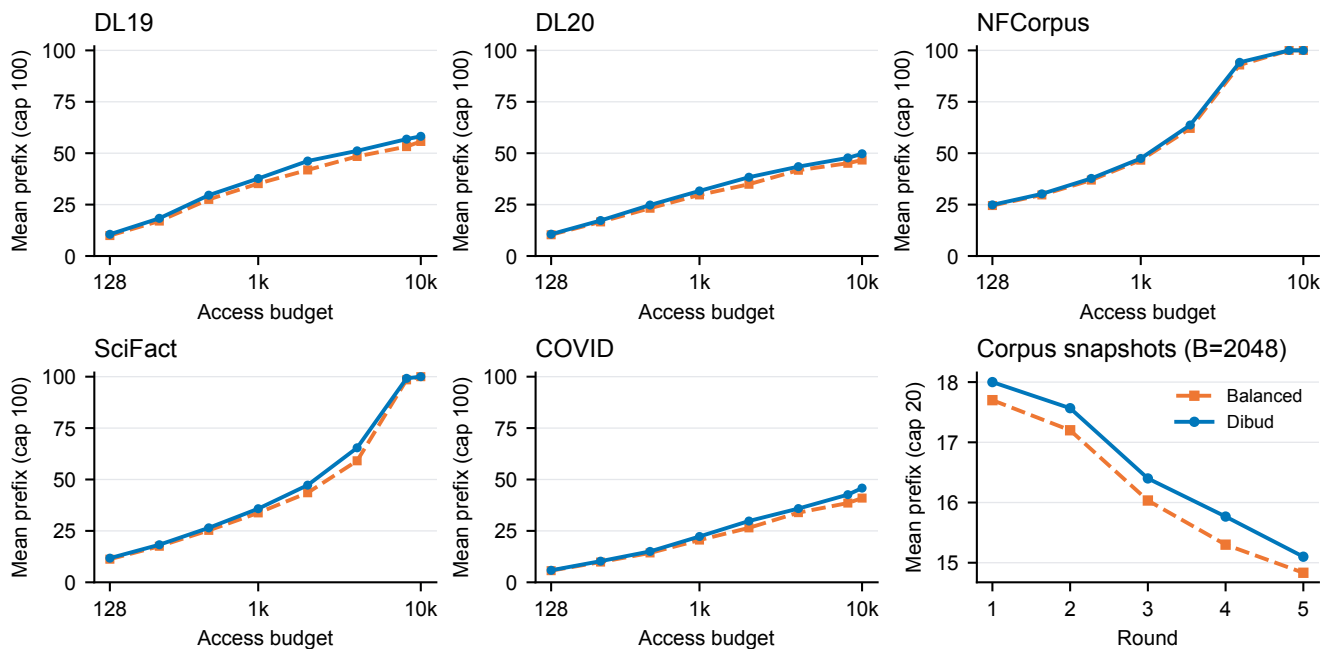


Fig. 2. Certified output under equal budgets. Static panels cap output at 100; the snapshot panel uses a cap of 20 and $B = 2048$. All plotted budget endpoints are observed.

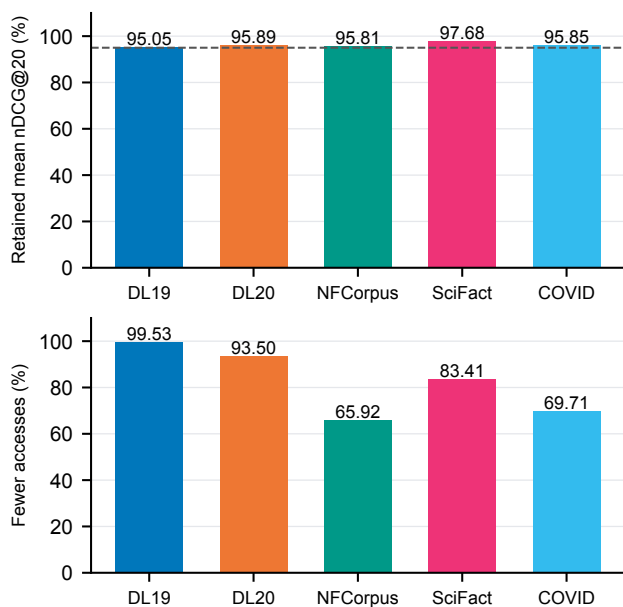


Fig. 3. Held-out mean quality retention and access reduction relative to exact Top-20 after 95% budget calibration. The dashed line marks the calibration target.

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